

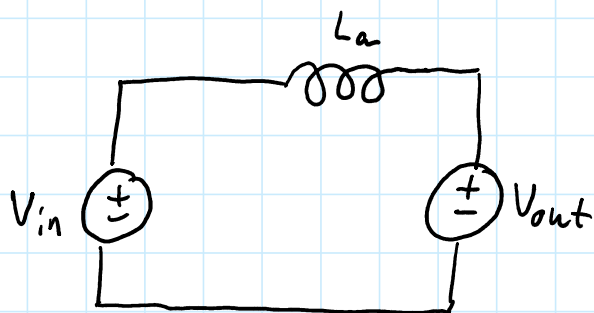
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onsdag 12. mars 2025 12:35

Problem 1:

a) $i_a(0) = I_0$, $i_a(t)$ for $0 < t < t_{on}$?

Switch is on: simplify diagram



Current through inductor:

$$i(t) = \frac{1}{L} \int_0^t V(\tau) d\tau + i_a(0)$$

$$= \frac{1}{L} \int_0^t V_{in} - V_{out} d\tau + I_0$$

$$= \frac{1}{L_a} \cdot (V_{in} - V_{out})t + I_0$$

$i_a(t_{on})$:

$$t_{on} = DT_s \Rightarrow i_a(t_{on}) = \frac{(V_{in} - V_{out})DT_s}{L_a} + I_0$$

b) $i_a(t)$ for $t_{on} < t < T_s$:

$$i_a(t) = \frac{1}{L_a} \int_{t_{on}}^t -V_{out} d\tau + i_a(t_{on})$$

$$= -\frac{V_{out}}{L_a} (t - t_{on}) + \frac{(V_{in} - V_{out})DT_s}{L_a} + I_0$$

$$i_a(T_s) = \frac{-V_{out}(T_s - DT_s) + V_{in}DT_s - V_{out}DT_s}{L_a} + I_o$$

$$= \frac{-V_{out}T_s + V_{in}DT_s}{L_a} + I_o = \frac{T_s(V_{in}D - V_{out})}{L_a} + I_o$$

$$c) \hat{i}_a(\bar{T}_s) = i_a(0)$$

$$\frac{T_s(DV_{in} - V_{out})}{L_a} + i(0) = i(0)$$

$$\Rightarrow DV_{in} = V_{out}$$

$$\text{Turnover ratio } \frac{V_{out}}{V_{in}} = D$$

It is step-down because $D \leq 1$

$$d) I_{mean} = \frac{1}{T_s} \int_0^{T_s} i_a(t) dt = \frac{1}{T_s} \left(\int_0^{t_{on}} i_a(t) dt + \int_{t_{on}}^{T_s} i_a(t) dt \right)$$

Can use this formula instead:

$$I_{mean} = I_o + \frac{1}{2}(\text{peak} - I_o) = I_o + \frac{1}{2}(i_a(t_{on}) - I_o)$$

$$= I_o + \frac{1}{2} \left(\frac{(V_{in} - V_{out})DT_s}{L_a} + I_o - I_o \right)$$

$$= \frac{(V_{in} - V_{out})DT_s}{2L_a} + I_o = \frac{(V_{in} - DV_{in})DT_s}{2L_a} + I_o$$

$$e) D_1 = 0,9 : V_{out} = 13,5V$$

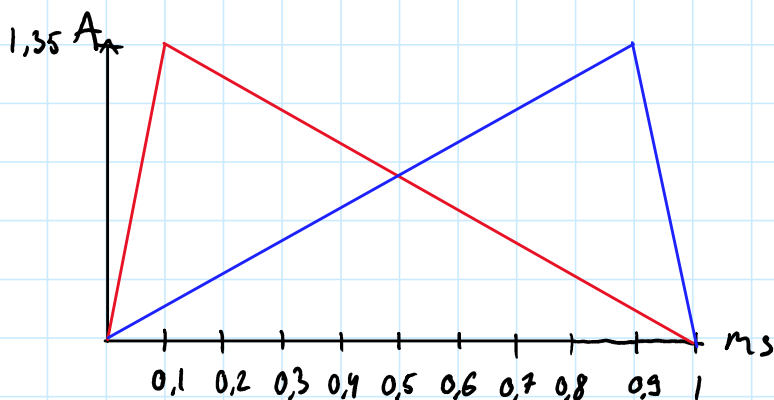
e) $D_1 = 0,9 : V_{out} = 13,5 V$
 $D = 0,1 : V_{out} = 1,5 V$

$i(t)$ for $0 < t < t_{on}$ and $t_{on} < t < T_s$ are both linear functions. We need to find $i(t_{on})$ to sketch $i(t)$ for one cycle

$$i(t_{on}) = \frac{(V_{in} - V_{out}) D T_s}{L_a} + I_o = \frac{(15V - 15 \cdot D V) D \cdot 1ms}{1mH}$$

$$= (15 - 15D) D A$$

$D_1 : i(t_{on}) = 1,35 A \quad D_2 : 1,35$



Problem 2:

- a) The rectifier will output different DC voltages because the AC voltages can be in the range 100 to 230V. The buck converter just multiplies this DC voltage by a constant $D \in [0, 1]$. It will probably be wrong, so you need a duty cycle controller. You also need to change the duty cycle based on how full the battery is.

b) $D=1 \quad 100 \cdot \frac{1}{n} \cdot 0,9 \cdot 1 = 5 \Rightarrow n = 18$

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$$D=0,2 \quad 230 \cdot \frac{1}{n} \cdot 0,9 \cdot 0,2 = 4 \Rightarrow n = 10,35$$

$n = 10,35$ is the smallest ratio
and also works for the 100V case.

c) Part b:

$D=1$: 100V gives $n = 18$

Lowest duty cycle when $V_s = 230V$

$$230 \cdot \frac{1}{18} \cdot 0,9 \cdot D = 4 \Rightarrow D = 0,347$$

$$d) \text{ Active power } P = U_{\text{batt}} I_{\text{batt}} + P_{\text{loss}}$$

$$= 10W + 5W = 15W$$

$$S = \frac{P}{\cos \varphi} = \frac{15W}{0,95} = 15,79 VA$$

$$S = V_s I_s \Rightarrow I_s = \frac{S}{V_s} = \frac{15,79 VA}{230V} = 0,068 A$$

Problem 3:

$$a) V_o = \frac{1}{1-D} V_s \quad \text{and} \quad \frac{1}{1-D} > 1, \quad \text{so}$$

V_o must be bigger than V_s .

$$b) V_L = L \frac{di}{dt}$$

$$i(t) = \frac{1}{L} \int_0^t v(\tau) d\tau + i_a(0)$$

If the mean voltage over the inductor
was $\neq 0$ then the current would increase/decrease
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$$c) V_s t_{on} + (V_s - V_o) t_{off} = 0$$

$$t_{on} = DT_s \quad \text{and} \quad t_{off} = T_s - DT_s$$

$$V_s DT_s + (V_s - V_o)(T_s - DT_s) = 0 \quad | \cdot \frac{1}{T_s}$$

$$\cancel{V_s D} + V_s - V_o - \cancel{DV_s} + DV_o = 0$$

$$V_s - V_o + DV_o = 0$$

$$V_s + V_o(-1 + D) = 0$$

$$\frac{V_o}{V_s} = \frac{1}{1-D}$$